Domestic vs. Foreign Law: Portfolio Dynamics of Sovereign Debt

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Contents

1 Introduction **3**

2 Empirical Evidence **5**

2.1 Data . . . 5
2.2 Every government borrows under both laws . . . 6
2.3 Foreign-law debt is issued at longer duration . . . 7
2.4 The yield gap rises with the probability of default . . . 8
2.5 Issuing under foreign law costs a fee proportional to face value . . . 11

3 Model **13**

3.1 Small open Economy . . . 13
3.2 Timing . . . 14
3.3 Recursive Formulations . . . 14
3.4 Prices . . . 16
3.5 Equilibrium . . . 17

4 Quantitative Analysis **17**

4.1 Calibration . . . 17
4.2 Model and data . . . 21
4.3 Experiments: the dilution–default trade-off . . . 22

5 The Cost of Weaker Domestic Legal Protection **23**

5.1 Why governments do not issue short foreign-law bonds . . . 23
5.2 The gain from equal protection under both laws . . . 25

A Appendix **31**

A.1 Shares of outstanding debt . . . 31
A.2 Duration differences . . . 32
A.3 A simple model of the yield gap . . . 33
A.4 Depreciation at default . . . 34
A.5 Instruments and curves . . . 35
A.6 Stationarity, cointegration and inference . . . 35
A.7 Robustness of the yield-gap estimates . . . 35
A.8 Binned scatters by country and bond-by-bond estimates . . . 35
A.9 Placebo: Peru's Global Depositary Notes . . . 35
A.10 The cost of issuing . . . 35

A.11 How often and how much governments issue 38
A.12 Recovered debt in the numerical solution 38
A.13 The slope we target 39
A.14 Robustness to the reentry probability 40
A.15 The two experiments 40
A.16 The fee and the cost of borrowing 40
A.17 The fee ladder 43
A.18 Welfare: measurement and decomposition 43

1 Introduction

When a government defaults, its creditors cannot seize assets within its borders, and what they can seize abroad generally falls short of what they are owed. What they recover is instead negotiated with the government, which remains excluded from financial markets until the two sides settle (Eaton and Gersovitz, 1981; Aguiar and Gopinath, 2006; Arellano, 2008; Yue, 2010). The bonds' terms set the rules of that negotiation; for example, they may set the majority of bondholders that must accept a deal for it to bind all of them. Whether the government can change those terms depends on the bond's governing law, the law under which they are written and enforced. Under domestic law, the government can change them as it would any other law, even after the bonds have been sold. Under foreign law, a court outside the government's reach enforces them as written at issuance. In its 2012 restructuring, Greece used this difference to gain an advantage in the negotiation: it changed the terms of its Greek-law bonds but could not do so for its foreign-law bonds, and foreign-law bondholders recovered more on average (Zettelmeyer et al., 2013).

This paper studies how governments borrow under domestic and foreign law, empirically and quantitatively. The central object is how much investors expect to recover under each law. Estimates from past restructurings vary widely across countries and episodes (Cruces and Trebesch, 2013), and say little about what investors expect from a government that has not defaulted recently. We therefore develop a methodology that infers the expected recovery under each law from a country's own bond prices and the amount of debt it issues under each law, before any default occurs. We apply it to Colombia and find that investors expect to recover 40 percent of their claims under domestic law and 47 percent under foreign law. We then explain why Colombia issues under both laws, and measure what it would gain if its domestic-law debt offered investors the same protection in default as its foreign-law debt.

We begin by documenting how emerging-market governments use the two laws. We combine debt-office records with bond-level data from Dealogic on every bond twelve governments issued between 2006 and 2024, and with daily bond prices. Three facts emerge. First, all twelve governments issue under both laws, with foreign-law shares of marketable debt between 5 percent in Brazil and 63 percent in Uruguay. Second, foreign-law debt is issued at longer duration in every country. Third, the yield gap between the two laws widens with default risk. To measure it, we price each foreign-law bond's payments on the domestic-law yield curve, following Chamon et al. (2018), and use bonds that Brazil, Colombia, Paraguay and Russia issue in their own currency under foreign law,

so that the two bonds differ only in their governing law. In Colombia, the gap rises by 23 basis points for every 100 basis points of CDS.

We then develop a quantitative model of sovereign default in which a government that lacks commitment chooses how much to borrow under each law (Arellano, 2008; Chatterjee and Eyigungor, 2012). Debt under each law has its own recovery and its own duration, as in models where a government chooses among several debt instruments (Arellano and Ramanarayanan, 2012; Passadore et al., 2026). Calibrated to Colombia, with durations of 3.75 years under domestic law and 8.71 years under foreign law, the model needs two moments to pin down the two recoveries: the slope of the yield gap and the share of debt under foreign law.

The model explains why a government that lacks commitment issues under both laws. The government can commit neither to repay nor to a path of future borrowing. On the one hand, foreign-law debt recovers more, which protects investors against default. On the other hand, domestic-law debt is shorter, which protects them against dilution: new borrowing lowers the value of debt that is still outstanding, and long debt stays exposed for longer (Hatchondo et al., 2016; Aguiar et al., 2019). Each law addresses one of the two commitment problems, so the government uses both. With equal durations, it issues only the high-recovery bond; with equal recoveries, only the shorter bond.

We then ask why governments do not issue short foreign-law bonds, which would combine the higher recovery of foreign law with protection against dilution. Foreign-law bonds are placed through bank syndicates for a fee that is a fixed share of face value, independent of the bond's maturity and of the amount issued (Section 2.5), while domestic-law bonds are auctioned at almost no cost. The fee is a tax on rolling over debt, so it weighs most on short bonds. With only one-period bonds, a fee the size of the one we measure barely changes the government's choice. With a long foreign-law bond available, the same fee reduces short foreign-law debt from 6.7 to 0.5 percent of GDP, because the government can offer investors the recovery of foreign law through the long bond, on which it pays the fee less often.

Last, we use the model to quantify the cost of weaker legal protection under domestic law. We consider a counterfactual in which domestic-law debt offers investors the same recovery as foreign-law debt, 47 percent instead of 40. In this counterfactual, welfare rises by 8.2 basis points of permanent consumption. That is about 12 percent of the gain the baseline model generates when aggregate risk and default costs are removed. The government borrows more, by 0.4 percentage points of GDP, defaults less often, by 0.3

percentage points a year, and spreads fall by 46 basis points on domestic-law debt and 22 basis points on foreign-law debt. Decomposing the gain, we find that most of it comes from changes in the government's borrowing and default policies, and a smaller part from lower borrowing costs, the channel the literature on governing law emphasizes (Chamon et al., 2018). The result is robust to positive initial levels of debt. As is standard in quantitative models of sovereign default, the government is more impatient than investors. Its impatience may reflect political frictions more than the preferences of households (Aguiar et al., 2020). We therefore also evaluate the gain with a higher household discount factor, and households more patient than the government also gain.

Related literature.

2 Empirical Evidence

2.1 Data

Our sample covers the central governments of twelve emerging markets from 2006 to 2024: Brazil, Chile, Colombia, Hungary, Mexico, Paraguay, Peru, Poland, Russia, South Africa, Turkey and Uruguay. We study their marketable debt: central-government bonds that any private investor can buy and later resell, net of the central bank's holdings. This definition excludes four kinds of public debt. The first is loans from multilateral and bilateral lenders. The second is bills issued by the central bank and the debt of state companies. The third is retail savings bonds. The fourth is bonds sold directly to public institutions, such as state pension funds or development banks, without an auction. These institutions are often required by law to hold government debt, a form of financial repression (Reinhart and Sbrancia, 2015; Chari et al., 2020), so their purchases do not reflect the price at which private investors would lend. Bonds that public institutions buy at the government's regular auctions remain in the sample.

Domestic-law bonds come from each debt office's records, which report for every placement its date, amount, coupon, maturity and yield at issuance. Foreign-law bonds come from Dealogic's bond-level issuance data, which report the same variables together with the governing law. On both sides we keep only bonds the central government itself issues, and we treat each re-tap of an outstanding bond as a separate placement. Daily bond prices come from Bloomberg, official sources and Capital IQ Pro. When more than one source covers a bond, we use the source with the widest coverage and check the results against the others. Daily five-year CDS spreads come from Markit, through Capital

IQ Pro. Underwriting fees come from Dealogic, and the issuer's own expenses from the final prospectuses that governments file with the SEC. GDP in current US dollars comes from the World Bank's World Development Indicators.

2.2 Every government borrows under both laws

Table 1. Marketable Debt by Governing Law

	Domestic law		Foreign law		Total
	% GDP	% debt	% GDP	% debt	% GDP
	(1)	(2)	(3)	(4)	(5)
Brazil	41.9	95.0	2.2	5.0	44.1
Chile	14.5	76.2	4.7	23.8	19.2
Colombia	14.3	70.3	6.1	29.7	20.4
Hungary	33.6	69.8	14.7	30.2	48.3
Mexico	23.5	84.0	4.5	16.0	28.0
Paraguay	1.2	48.1	5.6	51.9	6.8
Peru	10.2	65.3	5.2	34.7	15.4
Poland	29.6	79.2	7.8	20.8	37.4
Russia	6.5	79.1	1.5	20.9	8.0
South Africa	41.4	91.5	3.9	8.5	45.3
Turkey	19.2	75.5	5.5	24.5	24.7
Uruguay	11.8	37.5	20.1	62.5	31.9

Notes: Marketable central-government debt outstanding at each 31 December, averaged over 2006–2024. Marketable debt consists of the central-government bonds that any private investor can buy on the same terms and later resell, net of the central bank's holdings. Columns (1) and (3) report the domestic-law and foreign-law stocks as a percentage of GDP, columns (2) and (4) as a percentage of total marketable debt, and column (5) the total as a percentage of GDP. Debt and GDP are in current US dollars, GDP from the World Bank (NY.GDP.MKTP.CD). Domestic-law stocks come from each debt office's records and foreign-law stocks from Dealogic. Robustness checks on column (4) are in Table 9.

Every government in our sample borrows under both laws. Table 1 reports each government's marketable debt under each law, as a percentage of GDP and as a share of its total marketable debt, averaged over 2006–2024. Foreign-law bonds make up between 5 percent of marketable debt in Brazil and 63 percent in Uruguay, and between 20 and 35 percent in seven of the twelve countries. As a percentage of GDP, foreign-law debt ranges from 1.5 percent in Russia to 20 percent in Uruguay, and total marketable debt from 7 percent in Paraguay to 48 percent in Hungary.

We measure the stock of marketable debt under each law at the end of each year from

2006 to 2024. A bond counts as outstanding from its settlement date until it matures. The domestic-law stock comes from the debt offices' records, with inflation-indexed bonds valued at what the government owes on them, and we convert it to dollars at the year's average exchange rate. The foreign-law stock is the dollar value of each bond at issuance, from Dealogic. Appendix A.1 reports the shares under other exchange-rate conventions, with foreign-law bonds revalued each year, and without the holdings of all public institutions.

2.3 Foreign-law debt is issued at longer duration

Table 2. Duration at Issuance by Governing Law

	Domestic law				Foreign law				Gap	
	Median (1)	Mean (2)	Face (3)	GDP (4)	Median (5)	Mean (6)	Face (7)	GDP (8)	Median (9)	Face (10)
Brazil	3.47	4.35	2.34	2.28	8.40	9.65	9.87	9.82	4.93	7.53
Chile	8.85	10.31	8.36	8.57	9.90	12.13	12.35	12.26	1.05	3.99
Colombia	3.75	4.74	5.79	5.52	8.71	10.56	10.58	10.62	4.96	4.79
Hungary	2.84	3.74	2.46	2.35	7.54	7.60	8.45	8.39	4.70	5.99
Mexico	0.48	2.67	2.00	1.96	8.87	10.14	11.12	11.13	8.39	9.12
Paraguay	3.46	3.88	4.08	3.87	8.74	10.54	10.51	10.54	5.28	6.43
Peru	8.64	9.28	9.10	9.30	11.77	13.46	13.26	12.97	3.13	4.16
Poland	2.56	3.97	3.72	3.63	8.19	8.75	8.80	8.80	5.63	5.08
Russia	5.80	5.49	4.54	4.64	8.00	8.98	9.51	9.55	2.20	4.97
South Africa	6.88	6.89	3.57	3.59	8.79	10.12	10.23	10.16	1.91	6.66
Turkey	1.43	2.25	2.13	1.86	7.64	7.83	7.43	7.57	6.21	5.30
Uruguay	8.19	8.55	8.91	8.78	9.45	10.55	11.51	11.47	1.26	2.60

Notes: Macaulay duration of each placement at the moment of sale, in years. Median and Mean count every placement once; Face weights each placement by the dollars it raised; GDP weights it by those dollars over its own year's GDP. Gap is foreign law minus domestic law. Samples start in 2004 or 2006 and end in 2024. Maturity at issuance, a common-window check and the governing law of foreign-law placements are in Tables 10, 11 and 12.

Foreign-law debt is issued at longer duration than domestic-law debt in every country in our sample. Table 2 reports the duration at issuance of each government's placements under each law. The median foreign-law placement has a duration between 7.5 years in Hungary and 11.8 years in Peru, and the median domestic-law placement between 0.5 years in Mexico and 8.9 years in Chile. The gap between the two medians ranges from 1.1 years in Chile to 8.4 years in Mexico.

We measure the duration of every placement of marketable debt at the time of sale, following Macaulay (1938): the average of the years until each of the bond's payments, each

weighted by its discounted value relative to the bond's price. For a bond with face value of 1 that pays coupon c at $n_1, \ldots, n_J$ years after settlement and is sold at yield y,

$$d = \frac{1}{p}\left[\sum_{j=1}^{J} \frac{n_j\, c}{(1+y)^{n_j}} + \frac{n_J}{(1+y)^{n_J}}\right], \qquad p = \sum_{j=1}^{J} \frac{c}{(1+y)^{n_j}} + \frac{1}{(1+y)^{n_J}}.$$

As in Arellano and Ramanarayanan (2012), we use duration because it is comparable across bonds with different coupons. A re-tap of an outstanding bond counts as a separate placement, measured from its own settlement date. For each law we report the median and mean across placements, and two weighted means: one weights each placement by the dollars it raised, and the other by those dollars over its year's GDP. Appendix A.2 describes how we treat each type of bond, and reports maturity at issuance, the medians over a common 2006–2024 window, and the laws under which the foreign-law placements are written.

2.4 The yield gap rises with the probability of default

The yield gap between domestic-law and foreign-law bonds of the same government rises with the probability of default. We define the gap as the domestic-law yield minus the foreign-law yield. If investors expect to recover more under foreign law, that extra recovery is worth more when default is more likely, so the gap widens as default becomes more likely. If investors do not care which court enforces the bond, the gap reflects only the other characteristics of the two bonds and does not move with default risk (Appendix A.3 derives this relation in a simple pricing model). The yields of a government's domestic-law and foreign-law bonds cannot be compared directly, because the bonds differ in duration (Section 2.3), coupon and, in most cases, currency. We therefore use the foreign-law bonds that Brazil, Colombia, Paraguay and Russia issue in their own currency (Table 3). Following Chamon et al. (2018), we construct for each of them a domestic-law bond with the same payments. Chamon et al. (2018) build this bond from currency forwards, which requires that the currency not depreciate in default, an assumption that fits the euro area but not emerging markets, where defaults coincide with large depreciations (Reinhart, 2002; Na et al., 2018). Comparing bonds in the same currency avoids it, and gives the first estimates of the value of governing law for Brazil, Colombia and Paraguay.

The construction proceeds in four steps. A foreign-law bond has a daily price $q_{F,t}^{LC}$ and payments $c_1, \ldots, c_n$ per unit of face on dates $s_1 < \cdots < s_n$, the last one including princi-

Table 3. Local-Currency Foreign-Law Bonds

Country	Law	Currency	Bonds	Instruments	Window
Brazil	New York	BRL	3	Global BRL 12.5% '22, 10.25% '28, 8.5% '24	2014–2020
Colombia	New York	COP	3	Global TES 9.85% '27, 7.75% '21, 4.375% '23	2017–2020
Paraguay	New York	PYG	3	Global 7.9% '31, 8.5% '35, 8.5% '38	2025–2026
Russia	English	RUB	1	7.85% '18 eurobond	2014–2017

Notes: All bonds are nominal, fixed-coupon and non-callable. Brazil's and Colombia's windows end at the COVID break, Russia's when its bond falls below one year to maturity, and Paraguay's at the end of available data.

pal. First, we compute its yield to maturity $y_{F,t}$ from its price,

$$q_{F,t}^{LC} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + y_{F,t}\right)^{s_k - t}}. \tag{1}$$

Second, we fit the domestic-law zero-coupon yield curve $z_t(\cdot)$ on day t, and discount the same payments on it,

$$\widetilde{q}_{D,t}^{LC} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + z_t(s_k - t)\right)^{s_k - t}}. \tag{2}$$

Third, we compute the yield $\widetilde{y}_{D,t}$ that this price implies,

$$\widetilde{q}_{D,t}^{LC} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + \widetilde{y}_{D,t}\right)^{s_k - t}}. \tag{3}$$

Fourth, the yield gap is $\widetilde{y}_{D,t} - y_{F,t}$. The two bonds then share issuer, currency and payments, and differ only in their governing law. The domestic-law curve has to contain only domestic-law bonds. We use published curves where they do: Bloomberg's for Brazil and Colombia and the central bank's for Russia. For Paraguay, where no such curve exists, we fit a Nelson–Siegel curve each day to the domestic-law bonds quoted that day (Appendix A.5).

We measure the probability of default with the government's five-year CDS spread, and estimate how the gap moves with it, country by country, in first differences:

$$\Delta\left(\widetilde{y}_{D,i,t} - y_{F,i,t}\right) = \gamma\, \Delta \mathrm{CDS}_t + \Gamma' X_{i,t} + \mu_i + u_{i,t}, \tag{4}$$

where i indexes bonds and t trading days. The controls $X_{i,t}$ are the change in the bond's bid-ask spread relative to the domestic benchmark, which accounts for differences in trading costs; its time to maturity; and the change in the VIX, which separates movements in the CDS spread that reflect global risk aversion. The μ_i are bond fixed effects. The co-

efficient γ is the number of basis points the gap moves per basis point of CDS. We use first differences because we cannot reject a unit root in the gap or the CDS spread (Appendix A.6), which also removes any constant wedge between the two bonds, such as an error in fitting the curve. We use Driscoll–Kraay standard errors because the CDS spread is common to all of a country's bonds.

Table 4. The Yield Gap and the CDS Spread

	Brazil	Colombia	Paraguay	Russia
Δ CDS spread	0.664***	0.227***	0.157**	0.076***
	(0.091)	(0.050)	(0.072)	(0.023)
	[0.063]	[0.038]	[0.078]	[0.030]
Liquidity, maturity controls	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	—
Bonds	3	3	3	1
Trading days	1,490	573	330	656
Observations	3,399	1,373	713	653
R^2	0.096	0.038	0.042	0.072
Window	2014–2020	2017–2020	2025–2026	2014–2017

Notes: Estimates of γ in equation (4), country by country. The dependent variable is the daily change in the yield gap, in basis points; ΔCDS is the daily change in the five-year CDS spread. Controls: the change in the bond's bid-ask spread relative to the domestic benchmark, time to maturity and the change in the VIX, with bond fixed effects. Driscoll–Kraay standard errors in parentheses (Newey–West for Russia, which has one bond), Huber–White in brackets. Stars follow Driscoll–Kraay: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The gap rises with default risk in every country (Table 4). For every 100 basis point rise in the CDS spread, it rises by 66 basis points in Brazil, 23 in Colombia, 16 in Paraguay and 8 in Russia, and each estimate is significant at the 5 percent level or better. These slopes are lower bounds on the difference in recoveries a dollar investor expects, because the currency jumps in default and lowers the dollar value of both recoveries. Measuring that jump with CDS contracts settled in dollars and in pesos, Colombia's slope rises from 0.23 to 0.36 (Appendix A.4).

Reading the gap as the value of governing law requires that the same investors hold both bonds. Each country's sample therefore starts once nonresidents can buy, hold and sell the domestic-law bond without restrictions or punitive taxes, and nonresidents held between 16 and 28 percent of the domestic-law market in Brazil, Colombia and Russia over these windows. As a placebo, we apply the same construction to Peru's sol Global Depositary Notes, which are held offshore like the other bonds but wrap domestic-law bonds, so the

governing law is the same on both sides; their gap does not move with Peru's CDS spread (Appendix A.9). Appendix A.7 reports estimates net of Colombia's withholding tax and over alternative windows.

2.5 Issuing under foreign law costs a fee proportional to face value

Issuing under foreign law costs the government a fee proportional to the face value of the bonds it sells, at a rate that does not depend on the maturity of the bonds or on the amount issued. A government that sells a foreign-law bond hires a syndicate of banks, which buy the bonds and place them with investors, and the banks keep a share of face value, the underwriting fee. Domestic-law bonds are sold at auctions run by the central bank or the debt office, which charge no comparable fee. Because the fee is paid each time a bond is issued, a government that borrows short under foreign law pays it more often for the same debt. Section 5.1 uses this fact to explain why governments do not issue short foreign-law bonds.

We measure the fee with Dealogic's gross spread, the underwriting fee of each bond in basis points of face value. We restrict the sample to bonds issued under New York law, for which Dealogic records a fee for 92 percent of the bonds, against 9 percent of English-law bonds and 43 percent of Japanese-law bonds. The sample has 368 bonds issued by ten governments between 2006 and 2024; Dealogic records no fee for Paraguay and Russia. Appendix A.10 repeats the analysis for the bonds issued under other foreign laws: bonds issued under English law in euros pay about the same fee as New York-law bonds, 14 against 15 basis points, while bonds issued in yen and Swiss francs pay more.

Each government pays its own rate (Table 5). The median fee ranges from 4 basis points in Chile and Peru to 25 in Brazil and Colombia. Figure 1 plots the fee of each bond against its maturity and against its size, after removing country and year averages.

The rate does not depend on the maturity of the bond or on the amount issued. Comparing bonds of the same country and year, a 30-year bond pays 0.1 basis points less than a 10-year bond, with a 95 percent confidence interval from -1.1 to $+0.9$ basis points, and maturity accounts for 0.1 percent of the variation in fees. A bond twice as large pays 0.3 basis points less, with an interval from -0.8 to $+0.2$, and size accounts for 3.4 percent of the variation. Each government therefore faces a single rate per unit of face value, whatever the size or maturity of the bonds it issues. In the corporate Eurobond market, the fee does not vary with the size of the issue either, but it rises with maturity (Kollo, 2005; Esho et al., 2006). Melnik and Nissim (2003) find no relation to maturity or size among 37

Table 5. Underwriting Fee on Foreign-Law Bonds

	Bonds	Median	Mean	Mean, face-weighted
	(1)	(2)	(3)	(4)
Brazil	50	25.0	24.4	22.9
Chile	47	4.0	5.6	5.3
Colombia	49	25.0	27.7	25.7
Hungary	10	18.0	19.1	18.2
Mexico	72	19.0	22.2	19.3
Peru	23	4.0	8.1	7.1
Poland	15	14.0	14.8	15.2
South Africa	19	13.0	13.8	13.7
Turkey	56	8.0	8.2	7.9
Uruguay	27	10.0	11.6	10.7

Notes: Underwriting fee (gross spread) in basis points of face value, from Dealogic, New York-law bonds, 2006–2024, one observation per bond, re-openings included. Columns (2) and (3) count every bond once; column (4) weights each bond by its face value in US dollars. Dealogic records no fee for Paraguay and Russia.

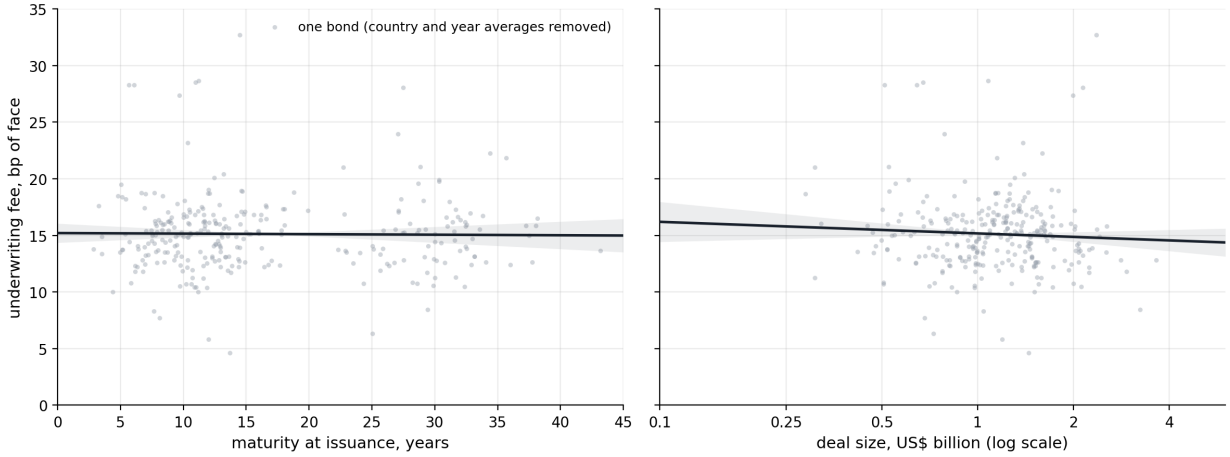


Figure 1. Underwriting Fee, Maturity and Size

Notes: Underwriting fee of each New York-law bond, in basis points of face value, against its maturity at issuance (left) and its face value in US dollars, on a log scale (right), 2006–2024, with country and year averages removed. The lines come from one regression of the fee on maturity and on the log of face value, with country and year effects. The sample excludes Mexico, four of whose bonds of US$0.5 billion or less pay 62 to 120 basis points (296 bonds).

sovereign and agency bonds issued in 1996–97. For the emerging-market governments in our sample, 2006–2024, the fee depends on neither.

The government also pays a flat amount for each deal, which is similar across countries and deals. These are its own expenses: lawyers, listing and paying agents, which the prospectuses filed with the SEC state as a single estimate. The median ranges from US$120,000 in Uruguay to US$305,000 in South Africa, a deal twice as large costs about 14 percent more in expenses, and expenses account for about 8 percent of the cost of a deal (Table 13).

Domestic auctions carry no comparable charge. In Brazil and Russia, the law states that the central bank runs them free of charge. Chile, Colombia, Turkey and Paraguay pay their central banks small flat amounts or reimburse their costs; in Chile, the fee for an issuance programme, relative to the size of the programme, is less than a thirtieth of the fee Chile pays on its foreign-law bonds. Uruguay is the exception: its central bank charges 1 basis point a year on the stock of domestic-law bonds, and twice that on foreign-law debt. For the other countries we find no charge in the documents we reviewed.

3 Model

In this section, we present a long-term debt model of sovereign default with two assets and heterogeneous recovery rates after default. Section 3.1 describes the macroeconomic environment, section 3.2 describes the timing, 3.3 characterizes the decisions of the government given prices, section 3.4 defines bond prices and section 3.5 defines the equilibrium.

3.1 Small open Economy

Time is discrete and denoted by $t \in \{0, 1, 2, \ldots\}$. The small open economy receives a stochastic stream of income denoted by y_t. Income follows a first-order Markov process $\mathbb{P}\left(y_{t+1} = y' \mid y_t = y\right)$. The government is benevolent and trades foreign-law and domestic-law bonds to smooth the household's consumption. The household evaluates consumption streams, c_t, according to:

$$\mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t u\left(c_t\right)\right],$$

with time-preference $\beta \in (0, 1)$ and utility function $u(\cdot)$, with $u'(\cdot) > 0$ and $u''(\cdot) < 0$.

The government issues foreign-law and domestic-law debt when it is not under default. As in Chatterjee and Eyigungor (2012) and Hatchondo and Martinez (2009) each unit of domestic (foreign) law debt matures with probability λ_D (λ_F). Bonds are issued at prices $q^{ND}_{D,t}$ and $q^{ND}_{F,t}$ at period t, and in equilibrium, those prices will depend on the current income and the vector of the next period's bond position $(y, b_{D,t+1}, b_{F,t+1})$. The government's budget constraint is:

$$c_t + \lambda_D b_{D,t} + \lambda_F b_{F,t} = y_t + q_{F,t}[b_{F,t+1} - (1-\lambda_F)b_{F,t}] + q_{D,t}[b_{D,t+1} - (1-\lambda_D)b_{D,t}],$$

where $\lambda_D b_{D,t}$ ($\lambda_F b_{F,t}$) are the total principal payments due corresponding to domestic-law (foreign-law) debt and $q_{j,t}[b_{j,t+1} - (1-\lambda_j)b_{j,t}]$ are the corresponding new debt issuances for debt under jurisdiction $j = D, F$.

3.2 Timing

The timing for the government is as follows and is summarized in Figure (2). There is limited enforcement of debt; thus, the sovereign defaults if it is optimal. When in default, the government loses access to the international lending market and incurs an output cost of $\phi(y)$ for each period it remains excluded. Consumption under exclusion c^D is exogenously determined by the budget constraint. During each period under exclusion, the government might regain access to credit markets with probability θ.

In our model, debt does not erase after default. This means that we have to keep track of debt prices during the default episode since defaulted debt has a positive recovery value. In a given period, before θ has been realized, domestic-law defaulted bonds are valued according to q^D_D, while foreign-law bonds are valued according to q^D_F.

Let $b_{t,j}$ represent the total amount of debt defaulted in the past under legislation j. A fraction $(1-\nu_j)$ of defaulted debt under legislation j is written off when the government regains access to credit markets. Thus, after a favorable draw of θ the recovered outstanding debt for the next period under each legislation becomes $\nu_j b_{t,j}$. As in Passadore and Xu (2022), debt is not erased in default, and the government regains access with the recovered debt outstanding. Appendix A.12 describes how we implement recoveries.

Bondholders of defaulted debt of jurisdiction j receive a fraction $\nu_j < 1$ identical replacement bonds for each unit they hold during exclusion. After being allowed back to financial markets, the newly issued domestic-law bonds are evaluated at $q^{ND}_D(y, \nu_D b_{t,D}, \nu_F b_{t,F})$ while their foreign-law counterparts are evaluated at $q^{ND}_F(y, \nu_D b_{t,D}, \nu_F b_{t,F})$.

The government either defaults on both domestic-law and foreign-law debt or repays both. Defaults on the bonds of one law alone are rare: of the 19 bond restructurings between 2000 and 2018 that involved bonds under both laws, Erce et al. (2024) classify one as selective by governing law, and in the other cases where one class of bonds was spared, the line was drawn by who held the bonds and not by the law that governed them.

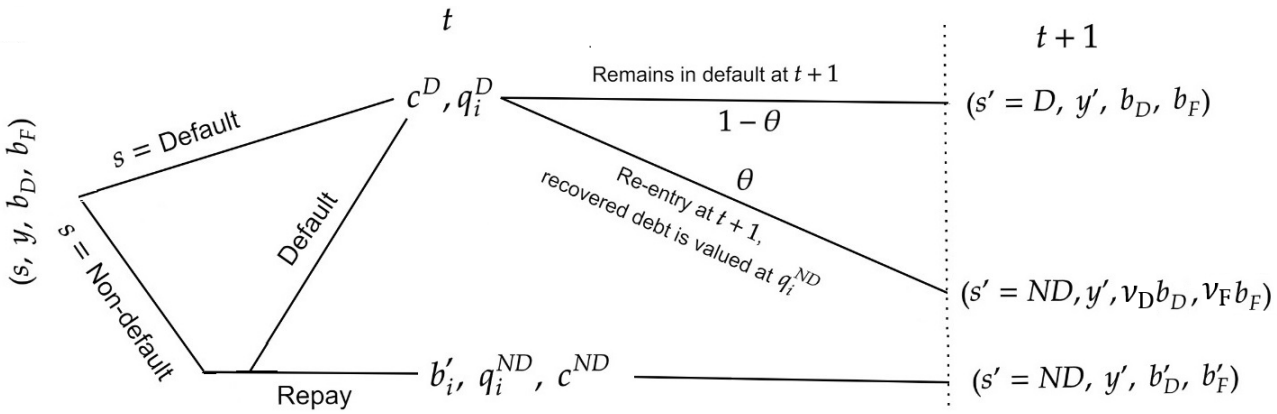


Figure 2. Timing

3.3 Recursive Formulations

Upon repayment, the government takes the price schedules as given and chooses a vector of bond holdings for the next period (b_D', b_F'). We look for a Markov equilibrium in which strategies are conditioned on (y, b_D, b_F). The unconditional value of the government is given by:

$$V(y, b_D, b_F) = \max_{d \in \{0,1\}} dV^D(y, b_D, b_F) + (1-d)V^{ND}(y, b_D, b_F). \tag{5}$$

Where the d is the government's default policy:

$$d(y, b_D, b_F) = \mathbb{1}_{\left\{V^D(y,b_D,b_F) > V^{ND}(y,b_D,b_F)\right\}}. \tag{6}$$

In case of default, there is no choice in the government's value:

$$V^D(y, b_D, b_F) = u\big(y - \phi(y)\big) + \beta \mathbb{E}_{y'|y}\left[\theta V^{ND}\left(y', \nu_D b_D, \nu_F b_F\right) + (1-\theta)V^D\left(y', b_D, b_F\right)\right] \tag{7}$$

Under repayment, the government chooses the vector of new issuances b_D', b_F':

$$V^{ND}(y, b_D, b_F) = \max_{b_D', b_F'} u(c) + \beta \mathbb{E}_{y'|y}\left[V^{ND}\left(y', b_D', b_F'\right)\right] \tag{8}$$

Subject to the budget constraint and a bound in the default probability induced by the choice of debt:

$$c + \lambda_D b_D + \lambda_F b_F = y + q_F^{ND}\left(y, b_D', b_F'\right)\left[b_F' - (1-\lambda_F)b_F\right] + q_D^{ND}\left(y, b_D', b_F'\right)\left[b_D' - (1-\lambda_D)b_D\right] \tag{9}$$

$$\delta\left(y, b_D, b_F\right) \equiv \mathbb{E}_{y'|y}\left[d\left(y', b_D', b_F'\right)\right] \leq \bar{\delta}. \tag{10}$$

As mentioned by Chatterjee and Eyigungor (2015) equation (10) is necessary for models with positive recovery rates to prevent the government from taking unrealistic amounts of debt right before defaulting. The solution to (8) subject to (9), and (10) generates the policy functions for foreign and domestic-law debt and an endogenous composition between the two.

3.4 Prices

The value of one unit of debt when the government is not in default is given by:

$$q_D^{ND}\left(y, b_D', b_F'\right) = \mathbb{E}_{y'|y}\left\{\left(1 - d\left(y', b_D', b_F'\right)\right)\frac{\lambda_D + (1-\lambda_D)q_D^{ND}\left(y', b_D'', b_F''\right)}{1+r} + d\left(y', b_D', b_F'\right)\frac{q_D^D\left(y', b_D', b_F'\right)}{1+r}\right\}, \tag{11}$$

$$q_F^{ND}\left(y, b_D', b_F'\right) = \mathbb{E}_{y'|y}\left\{\left(1 - d\left(y', b_D', b_F'\right)\right)\frac{\lambda_F + (1-\lambda_F)q_F^{ND}\left(y', b_D'', b_F''\right)}{1+r} + d\left(y', b_D', b_F'\right)\frac{q_F^D\left(y', b_D', b_F'\right)}{1+r}\right\}. \tag{12}$$

When not in default, the prices reflect the expected payoff of the lender, discounted under the risk-free rate. In particular, note that since debt lasts more than one period, the price incorporates the optimal behavior of the government in the future, which we denoted as b_j''. Conditional on repayment tomorrow, the lender gets the principal for the matured fraction of debt: $\lambda_j b_j$, plus tomorrow's price for the non-matured fraction of debt $(1 - \lambda_j)b_j$.

Note that in case of default, the price of one unit of debt is given by:

$$q_D^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\mathbb{E}_{y'|y}\left[q_D^D\left(y', b_D, b_F\right)\right] + \theta\nu_D q_D^{ND}\left(y, \nu_D b_D, \nu_F b_F\right), \tag{13}$$

$$q_F^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\mathbb{E}_{y'|y}\left[q_F^D\left(y', b_D, b_F\right)\right] + \theta\nu_F q_F^{ND}\left(y, \nu_D b_D, \nu_F b_F\right). \tag{14}$$

Consistent with Figure (2) defaulted debt prices reflect the probability of re-entering the market in the next period with probability θ. If the country receives a favorable draw, the recovered debt due is priced accordingly. Note that stated in this way, we allow for the possibility that the sovereign defaults on the recovered debt, a feature not present in models where debt gets erased upon re-entry. On the other hand, if the country remains excluded from financial markets $(1-\theta)$, it will face the same problem as today but possibly with a new income.

3.5 Equilibrium

Definition 1. *A Markov Equilibrium with state variables (y, b_D, b_F) is given by:*

(i) A set of policy functions for consumption $\hat{c}(y, b_D, b_F)$, default $\hat{d}(y, b_D, b_F)$, debt $\hat{b}'_D(y, b_D, b_F)$ and $\hat{b}'_F(y, b_D, b_F)$;

(ii) A set of value functions $\hat{V}(y, b_D, b_F)$, $\hat{V}^D(y, b_D, b_F)$, $\hat{V}^{ND}(y, b_D, b_F)$;

(iii) Price schedules $\hat{q}_D^D(y, b_D, b_F)$, $\hat{q}_F^D(y, b_D, b_F)$, $\hat{q}_D^{ND}(y, b'_D, b'_F)$, $\hat{q}_F^{ND}(y, b'_D, b'_F)$

such that:

1. *Given prices, the value functions solve: (5), (7), and (8), with associated policy functions;*
2. *Bond prices satisfy (11), (12), (13), (14).*

4 Quantitative Analysis

4.1 Calibration

We calibrate the model to Colombia at an annual frequency. Colombia offers the most complete data for our exercise: its debt office records every domestic-law placement, and

its foreign-law bonds are filed with the SEC, so the shares, durations, yield gap and fee we target all come from its own bonds. It is also a typical member of our sample. Its foreign-law share, 29 percent, lies within the range of seven of the twelve countries, and the slope of its yield gap lies between those of Russia and Brazil. Finally, Colombia has not defaulted since the 1930s, so the recoveries its investors expect cannot be read from its own restructurings and have to be inferred from prices, which is what our approach does. We solve a discretized version of the model with the algorithm of Gordon and Guerron-Quintana (2018), which extends Chatterjee and Eyigungor (2012) to models with several defaultable long-term bonds, building on the replication files of Guerrón-Quintana (2021) and Passadore and Xu (2022).

We set seven parameters outside the model. Household utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$, with relative risk aversion $\sigma = 2$, the standard value in the literature. Log output follows an AR(1),

$$\log y_t = \rho \log y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \tag{15}$$

which we estimate on Colombia's annual real GDP from the World Development Indicators over 1976–2024, after removing a linear trend, as in Arellano (2008) and Chatterjee and Eyigungor (2012). This gives $\rho = 0.77$ and $\sigma_\varepsilon = 0.026$. The risk-free rate is $r = 1.8$ percent a year in real terms, following Bocola et al. (2019). The reentry probability is $\theta = 0.25$, a mean exclusion of four years. Colombia offers no default from which to estimate it, and estimates of the length of exclusion after a default range from about three to six years, depending on the definition of market access and on the size of the haircut (Gelos et al., 2011; Cruces and Trebesch, 2013). Appendix A.14 reports the calibration under other values of θ. Coupons are normalized to zero, so each bond is a geometric-decay perpetuity whose Macaulay duration is $D = (1+r)/(\lambda+r)$. We set each λ so that this duration equals the median duration at issuance of Colombia's debt under that law (Section 2.3): 3.75 years for domestic law and 8.71 years for foreign law, which gives $\lambda_D = 0.254$ and $\lambda_F = 0.099$.

We choose the remaining five parameters jointly to match five moments of Colombia's data over 2006–2024: the discount factor β, the two recoveries ν_D and ν_F, and the two parameters of the output loss in default, which follows Chatterjee and Eyigungor (2012),

$$\phi(y) = \max\left\{0,\, d_0\, y + d_1\, y^2\right\}.$$

Three of them are the moments that quantitative models of sovereign default usually target (Arellano, 2008; Hatchondo and Martinez, 2009; Chatterjee and Eyigungor, 2012;

Arellano and Ramanarayanan, 2012): the mean and the volatility of the spread, 2.40 and 0.75 percent, from the J.P. Morgan EMBI Global Colombia, and the ratio of marketable debt to GDP, 20.9 percent (Section 2.2). The other two are new and discipline the recoveries: the foreign-law share of marketable debt, 29 percent (Section 2.2), and the slope of the yield gap on the CDS spread (Section 2.4). Nearly all of Colombia's foreign-law debt is issued under New York law, 98 percent of the face value it issued under foreign law (Table 12), and the EMBI spread, the yield gap and the issuance fee of Section 2.5 all come from New York-law bonds, so we read the foreign-law bond in the model as a New York-law bond and its spread as the model counterpart of the EMBI spread. Debt in the model is the face value of both bonds over output.

We target how the yield gap moves with default risk and leave its average level untargeted. The average difference between the yields of the two laws contains the value of governing law, but also every other difference between the two bonds that does not depend on default risk: their liquidity, the investors who hold them, their tax treatment, their inclusion in bond indices, and the error in fitting the domestic-law curve. The value of a higher recovery is different, because investors collect it only if the government defaults. It is worth little when default is unlikely and rises with the probability of default. The slope of the gap on the CDS spread isolates this component, and differencing removes the wedges that stay constant. Targeting the level would attribute all of these wedges to the recoveries. In the model, the gap is zero whenever $\nu_D = \nu_F$, and it rises with the probability of default at a rate that increases with $\nu_F - \nu_D$.

To compute the slope in the model, we price bonds that resemble those in the data. In the data, the gap is measured on a panel of bonds whose remaining maturity shortens every day, and the regression in Section 2.4 absorbs this with time to maturity and bond fixed effects. The model has one bond under each law, each with a constant duration. Along the simulated equilibrium path, using the solved default and borrowing policies, we therefore price synthetic bonds with constant average maturities $M \in \{1, 3, 5, 7, 10\}$ years and decay $\lambda_M = 1/M$. At each maturity we price a domestic-law and a foreign-law version, which differ only in their recovery. The yield gap at maturity M is the difference between their spreads,

$$s^D_{M,t} - s^F_{M,t} = \lambda_M \left(\frac{1}{q^D_{M,t}} - \frac{1}{q^F_{M,t}} \right), \tag{16}$$

and the model counterpart of the CDS spread is the spread on the five-year foreign-law

bond, $s^F_{5,t}$. The model slope γ is the estimate of

$$\Delta\left(s^D_{M,t} - s^F_{M,t}\right) = \gamma\, \Delta s^F_{5,t} + \alpha_M + \varepsilon_{M,t} \tag{17}$$

over the periods with market access. The synthetic bonds span the range of maturities in the data, the maturity fixed effect α_M plays the role of the bond fixed effect, and holding each maturity constant plays the role of the maturity control. The regression therefore uses the same variation as in the data: changes over time in the gap of a given bond. The model is real, so its slope measures the recovery difference without the currency effect discussed in Section 2.4; Appendix A.13 explains what this implies for the slope we target.

Table 6. Parameters and Targets

A. Externally set				
Parameter	Value	Description	Source	
σ	2	Relative risk aversion	Standard	
ρ	0.77	Autocorrelation of log output	WDI, 1976–2024	
σ_ε	0.026	Std. deviation of output shocks	WDI, 1976–2024	
r	0.018	Risk-free rate	Bocola et al. (2019)	
θ	0.25	Reentry probability	Four-year exclusion	
λ_D	0.254	Decay, domestic-law debt	Duration, 3.75 years	
λ_F	0.099	Decay, foreign-law debt	Duration, 8.71 years	
B. Internally calibrated				
Parameter	Value	Main target	Data	Model
β	0.920	Marketable debt / GDP (%)	20.9	21.0
d_0	-0.188	Mean spread (%)	2.40	2.34
d_1	0.236	Spread volatility (%)	0.75	1.02
ν_D	0.40	Foreign-law share (%)	29.0	29.0
ν_F	0.469	Slope of the yield gap, γ	0.23	0.23

Notes: Annual frequency. Colombia, 2006–2024. The parameters in panel B are chosen jointly to match the five moments; each is listed next to the moment it affects most. The spread is the J.P. Morgan EMBI Global Colombia spread, and its model counterpart is the spread on the foreign-law bond.

Table 6 pairs each parameter with the moment it affects most, although all five are chosen jointly. The mean spread mainly disciplines d_0, its volatility d_1, and the debt ratio β. The two recoveries are pinned down jointly by the two new moments, as Figure 3 shows. The slope of the yield gap rises with the recovery gap $\nu_F - \nu_D$ (panel a). Holding that gap fixed, the foreign-law share rises with the level of recovery (panel b): as both recoveries

rise, the government borrows more and defaults more often, so the same recovery gap applies in more states. The calibration delivers a recovery of 40 percent on domestic-law debt and 47 percent on foreign-law debt.

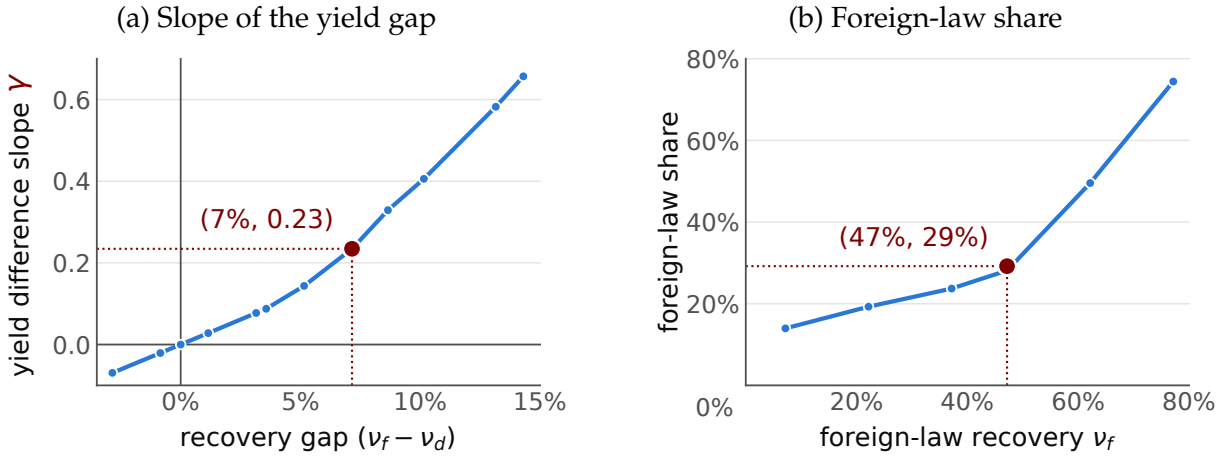


Figure 3. Identification of the Recoveries

Notes: Panel (a) plots the model slope of the yield gap on the CDS spread, γ, against the recovery gap $\nu_F - \nu_D$. Panel (b) plots the foreign-law share of debt against the foreign-law recovery ν_F, holding the recovery gap at its calibrated value. Other parameters at their calibrated values. The dots mark the calibration.

4.2 Model and data

Table 7. Untargeted Moments

	Model	Data
corr(spread, y)	-0.69	-0.12
corr(foreign-law share of issuance, y)	-0.69	-0.29
$\sigma(c)/\sigma(y)$	1.17	1.15
corr(c, y)	0.83	0.92
Debt service / GDP (%)	4.3	3.9
$\sigma(\log y)$	0.039	0.041
Default frequency (%)	3.7	—

Notes: Colombia, 2006–2024. Data moments for output and consumption use the cyclical component of logged series, HP filter with $\lambda = 100$, except $\sigma(\log y)$, which is implied by the linearly detrended AR(1); model moments use the simulated series. Model debt service includes amortization only, because coupons are zero; the data include coupons.

Investors in the calibrated model expect to recover 40 percent of their claims on domestic-law debt and 47 percent on foreign-law debt. As Table 6 shows, the model closely matches

the targeted moments, although it overstates the volatility of the spread, 1.02 percent against 0.75 percent in the data.

Table 7 compares the model with moments we do not target. The model reproduces the volatility of consumption relative to output, 1.17 against 1.15 in the data, and the burden of debt service, 4.3 percent of GDP against 3.9 percent, although debt service in the model excludes coupons. Spreads and the foreign-law share of new issuance are more countercyclical in the model than in the data: their correlations with output are -0.69 in the model, against -0.12 and -0.29 in the data. As in other models in which output is the only shock (Arellano, 2008), default risk in the model rises only when output falls, so spreads move more closely with output than they do in the data. The model implies that Colombia defaults in 3.7 percent of years, a frequency that, like the recoveries, follows from Colombia's own prices and debt.

The model also generates the split between the two laws that we observe: the government holds 29 percent of its debt under foreign law and the rest under domestic law. Section 4.3 shows that this split requires the two bonds to differ in both recovery and duration.

4.3 Experiments: the dilution–default trade-off

The government in the calibrated model borrows under both laws because each law addresses a different commitment problem. To show this, we run two experiments. In each, we remove one of the two differences between the bonds, keep every other parameter at its calibrated value, and solve the model again. In both, the government ends up using a single bond. Appendix A.15 reports the full results.

When the two bonds have the same duration, the government issues only the bond with the higher recovery. We give both bonds a common duration, on a grid from one to 8.71 years, and vary the two recoveries. At every duration, the government issues only the bond with the higher recovery, even when the recoveries differ by 0.02, and reversing which law recovers more reverses the result (Table 17). A higher recovery lowers the loss investors bear in default, so it lowers the compensation for default risk that the government pays on every unit of debt. With equal durations, nothing offsets this advantage.

When the two bonds have the same recovery, the government issues almost only the shorter bond. We give both bonds a common recovery, on a grid from 0 to 0.75, and keep the calibrated durations of 3.75 years for domestic law and 8.71 years for foreign law. The foreign-law share is between 0.1 and 0.3 percent at every recovery (Table 19). The government cannot commit to its future borrowing, and new borrowing lowers the price

of debt still outstanding. Long debt remains outstanding longer, so it is more exposed to this dilution, and investors pay less for it. The long bond keeps a role only when the short bond is much shorter: with a one-year domestic-law bond, the government holds between 48 and 68 percent of its debt in the long bond.

Each law therefore protects investors against one of the two risks. Foreign-law debt recovers more, so it protects them against default. Domestic-law debt is shorter, so it protects them against dilution. When the bonds differ in only one of these dimensions, the government uses only one bond. The split we observe arises because the bond with the higher recovery is also the longer bond, and Section 5.1 asks why the government does not issue a short bond with the higher recovery.

5 The Cost of Weaker Domestic Legal Protection

5.1 Why governments do not issue short foreign-law bonds

We now use the model to ask why Colombia does not issue a short foreign-law bond, which would protect investors against both default and dilution: it would carry the higher recovery of foreign law and the short duration of domestic-law debt. Between 2006 and 2024, Colombia's shortest foreign-law bond had a duration of 7.2 years at issuance, while its shortest domestic-law bond had a duration of 0.2 years. The fee documented in Section 2.5 is enough to explain this, but only because the government can offer investors the recovery of foreign law through a long bond. We add the fee to the model of Section 4 and run two experiments. In the first, the government chooses between a one-period domestic-law bond and a one-period foreign-law bond. In the second, we add a long foreign-law bond.

The government pays a fee f per unit of face value it issues under foreign law, and only when it issues. With foreign-law bonds collected in the set $\mathcal{F}$, the budget constraint becomes

$$c + \sum_j \lambda_j b_j \;=\; y + \sum_j q_j \left[b'_j - (1-\lambda_j) b_j \right] \;-\; f \sum_{j \in \mathcal{F}} \max\left\{0,\; b'_j - (1-\lambda_j) b_j \right\}. \tag{18}$$

A buyback pays no fee and earns no rebate. As in the data, the fee is a share of the face value issued, and the share does not depend on the maturity of the bond or on the amount issued. For Colombia we set $f = 0.0025$, the median fee of 25 basis points on its New York-law bonds (Table 5). The pricing equations are those of Section 3, one for

each bond, so the fee affects prices only through the government's default and borrowing policies.

The fee is a tax on rolling over debt. To keep one unit of face outstanding, the government issues each period the share λ that matures, and pays $f\lambda$ in fees. A one-period bond pays the fee on its whole face every period, while a long bond pays it on a fraction of its face. In terms of yields, the fee raises the cost of borrowing by approximately $\lambda f / q^2$, which is larger on the shorter bond (Appendix A.16).

With only one-period bonds, the fee does little. As in Section 4.3, without a fee the government borrows only under foreign law, because the two bonds are equally short and foreign-law debt recovers more. At 20 basis points it still holds 75 percent of its debt under foreign law, and it stops issuing under foreign law only at a fee of about 76 basis points. With only short bonds available, the only way to avoid the fee is to give up the higher recovery, and the government does so only when the fee is several times the one we measure.

Adding a long foreign-law bond changes the result. In the second experiment, the government chooses among a short domestic-law bond, a short foreign-law bond and a long foreign-law bond. The two foreign-law bonds share the recovery ν_F, and the domestic-law bond recovers ν_D. The short bonds last one period, $\lambda_D = \lambda_{Fs} = 1$, and the long bond has a duration of 2.32 years, $\lambda_{Fl} = (1+r)/2.32 - r = 0.420$, so that the ratio of the two durations matches Colombia's, $8.71/3.75 = 2.32$. These durations are shorter than in the data: we use one-period short bonds to keep the computation tractable, and we choose the long bond so that the ratio of the two durations matches Colombia's.[1] All other parameters keep their values from Section 4.1. Without a fee, the government holds short foreign-law debt worth 6.7 percent of GDP. At Colombia's fee of 25 basis points, this falls to 0.5 percent of GDP, a reduction of more than 90 percent, while total debt stays at 19 percent of GDP. Short foreign-law debt disappears at a fee of about 35 basis points, less than half the fee needed with only one-period bonds. A flat cost per issue of the size reported in the prospectuses, about US$300,000, leaves these results unchanged.

The difference between the two experiments is the long bond. It carries the same recovery as the short foreign-law bond, so the government can offer investors the protection of foreign law through it while paying the fee less often. Avoiding the fee then no longer means giving up the higher recovery: the government only moves its foreign-law borrowing to the long bond, and a fee the size of the one we measure is enough for it to do so. Its short

[1] With three bonds the state has four variables: output and three debt stocks. We solve the model on a grid of 25 points for output and 60 for each stock, $25 \times 60^3 \approx 5.4$ million states.

borrowing moves home, to the bond that protects investors against dilution but offers them less protection in default. Section 5.2 measures what Colombia loses because that bond recovers less than foreign-law debt.

5.2 The gain from equal protection under both laws

We now use the model to measure what Colombia loses because the bond that protects investors against dilution, its short domestic-law bond, offers them less protection in default than its foreign-law bond. We consider an economy in which the short domestic-law bond recovers 47 percent in default, as foreign-law debt does, instead of 40 percent, and keep every other parameter at its calibrated value. Higher protection under domestic law raises welfare by 8.2 basis points of permanent consumption. For comparison, eliminating aggregate risk and the costs of default in the baseline economy raises welfare by 71 basis points, so equal protection under the two laws delivers about 12 percent of that gain.

We measure welfare as the permanent percentage change in consumption Δ that makes the household indifferent between the baseline economy $\mathcal{E}_0$ and the economy with higher domestic-law recovery $\mathcal{E}_1$,

$$(1+\Delta)^{1-\sigma}\,\bar{V}^{\mathcal{E}_0} = \bar{V}^{\mathcal{E}_1}, \qquad \bar{V}^{\mathcal{E}} \equiv \sum_{y} \pi(y)\, V^{\mathcal{E}}(y,b), \tag{19}$$

where π is the invariant distribution of output and b is the debt at which we evaluate the gain. We evaluate it at two levels of debt: at zero debt, as in Chatterjee and Eyigungor (2012) and Hatchondo et al. (2016), and at the mean level of debt in the baseline economy. In the new economy we solve the model again, so the government's borrowing and default policies adjust to the higher recovery. At zero debt the gain is 8.2 basis points; at the mean level of debt it is 9.0.

Table 8 shows how the economy changes. With the same recovery under both laws, the government borrows more and defaults less often: debt rises by 0.4 percentage points of GDP, and the default frequency falls by 0.3 percentage points a year. Spreads fall by 46 basis points on domestic-law debt and by 22 basis points on foreign-law debt. In the baseline, the government offers investors protection against default by borrowing long under foreign law, which exposes that debt to more dilution. With the same recovery on its short domestic-law bond, it can offer investors that protection without lengthening its debt, and it moves its borrowing to the short bond.

A higher recovery affects welfare through three forces. First, investors expect to recover

Table 8. Raising the Recovery of Domestic-Law Debt from 0.40 to 0.47

A. Change in the economy	
Mean spread, domestic-law debt (bp)	−46
Mean spread, foreign-law debt (bp)	−22
Debt / GDP (percentage points)	+0.4
Default frequency (percentage points a year)	−0.3
B. Welfare gain (bp of permanent consumption)	
Total	+8.2
New prices, policies fixed	+2.1
New debt at reentry, policies fixed	−1.3
Both, policies fixed	+0.8
Change in borrowing and default policies	+7.4

Notes: Changes from the baseline economy, $\nu_D = 0.40$, to the economy with $\nu_D = 0.47$, model simulations. The welfare gain is Δ in equation (19), evaluated at zero debt. The change in policies is the total minus the gain with both changes and fixed policies. Appendix A.18 describes the decomposition.

more in default, so they pay more for the government's debt. Second, the government comes out of default owing more. Third, the government changes how much it borrows and when it defaults. To measure the first two, we keep the government's borrowing and default policies at their baseline values and change one feature at a time. Giving the government the prices of the new economy, with the baseline debt at reentry, raises welfare by 2.1 basis points. Giving it the larger debt at reentry of the new economy, with baseline prices, lowers welfare by 1.3 basis points. With both changes and fixed policies, welfare rises by 0.8 basis points, so the remaining 7.4 basis points of the total come from the government's response to the higher recovery. The literature on governing law emphasizes that stronger legal protection lowers a government's borrowing costs when default risk is high (Chamon et al., 2018). In our model this is the smaller part of the gain: most of it comes from a shift in the government's incentives to borrow and to default. Appendix A.18 describes these calculations.

Part of the gain comes from letting an impatient government borrow more. As is standard in quantitative models of sovereign default, the calibrated government discounts the future more than investors do, and this impatience may reflect political frictions more than the preferences of households (Aguiar et al., 2020). We check that the gain does not depend on this impatience. We evaluate it with a household discount factor β_H that ranges

from that of investors, $1/(1+r) = 0.982$, to that of the government, $\beta = 0.92$, holding the government's policies and the prices they imply fixed.

To show where the gain comes from, we follow Aguiar et al. (2020) and split the value of market access into three terms. The default-cost term is the welfare lost because output falls while the government is excluded; we obtain it by comparing the economy with one in which exclusion carries no output loss, but the government still comes out of default owing only the recovered share of its debt. The front-loading term is the value of bringing consumption forward by borrowing; we obtain it by comparing a household that receives for sure the mean consumption path of the economy with one that consumes mean output forever. An impatient household values front-loading, and a patient one does not. The variability term measures how much more or less volatile consumption is with market access than in autarky. The value of market access is the product of these three terms, and the welfare gain is the ratio of its values in the two economies, so each term contributes the ratio of its own values. Appendix A.18 gives the formulas.

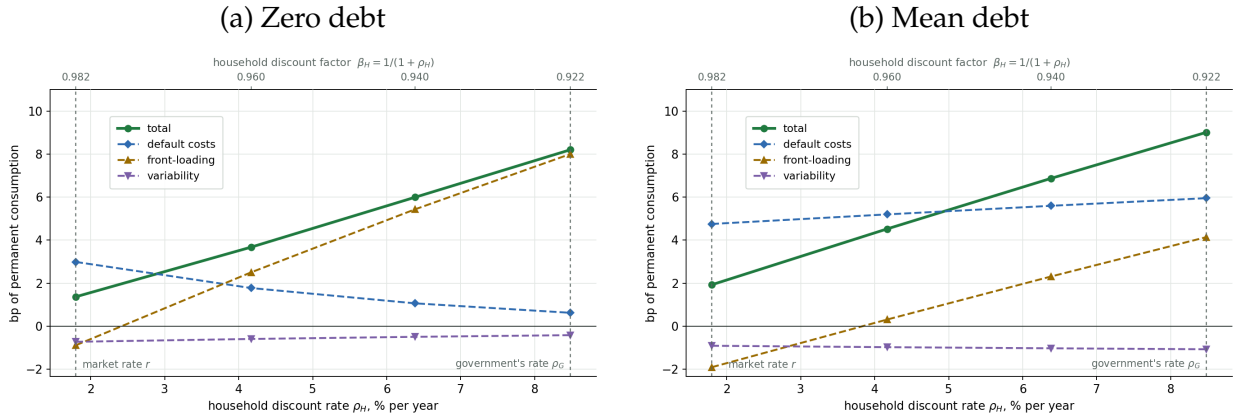


Figure 4. Welfare Gain and the Household's Discount Rate

Notes: Welfare gain from raising ν_D from 0.40 to 0.47, in basis points of permanent consumption, against the household's discount factor, from that of investors, $1/(1+r)$, to that of the government, β. The total is the product of the default-cost, variability and front-loading terms defined in Appendix A.18. Panel (a) evaluates the gain at zero debt and panel (b) at the mean level of debt in the baseline economy.

Figure 4 plots the gain and its three terms against the household's discount factor, at zero debt (panel a) and at the mean level of debt (panel b). Households gain at every discount factor between that of investors and that of the government. At zero debt, the total gain rises from 1.4 basis points for a household as patient as investors to 8.2 basis points for one as impatient as the government. The two ends differ in what drives the gain. A patient household values the lower costs of default, worth 3.0 basis points, while the additional borrowing lowers its welfare: its front-loading term is -0.9 basis points.

As the household becomes more impatient, it values the additional borrowing more; the front-loading line rises until it accounts for almost all of the gain at the government's discount factor, while the default-cost line falls to 0.6 basis points. The variability term is small and slightly negative throughout, because the government borrows more and consumption becomes slightly more volatile. At the mean level of debt, the lower costs of default matter more: they contribute between 4.7 and 6.0 basis points at every discount factor, and the total gain ranges from 1.9 to 9.0 basis points.

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A Appendix

A.1 Shares of outstanding debt

Inflation-indexed bonds enter the domestic-law stock at their accreted capital: the indexed amount times the current value of the index, which is what the government owes. We convert the domestic-law stock to dollars at the annual average of the year's daily exchange rates. GDP is converted at the same rate, so the exchange rate does not affect debt as a percentage of GDP. The foreign-law share does not involve GDP, so for it the 31 December rate would match the date at which both stocks are measured; Table 9 reports that alternative. A foreign-law tranche counts as foreign law unless its governing-law field in Dealogic names the home country, so a tranche with a blank field falls on the foreign side.

- **Dec. FX, column (2).** Converts the domestic stock at the 31 December exchange rate instead of the annual average.
- **Med. FX, column (3).** Converts the domestic stock at the annual median of daily exchange rates instead of the average.
- **Revalued, column (4).** Revalues each foreign bond at each year's exchange rate instead of keeping its dollar value on the issue date. Rates come from Dealogic's own dual-currency face values.
- **Net pub., column (5).** Also removes the holdings of public institutions that bought at competitive auctions.
- **Unresolved blanks (not in the table).** Excludes from the foreign side only the blank tranches whose law cannot be established another way.

Table 9. Foreign-Law Share of Marketable Debt: Robustness

	Headline (1)	Dec. FX (2)	Med. FX (3)	Revalued (4)	Net pub. (5)
Brazil	5.0	5.2	5.0	5.1	—
Chile	23.8	24.2	23.7	23.7	23.8
Colombia	29.7	30.2	29.4	29.8	31.6
Hungary	30.2	30.5	30.0	30.2	30.6
Mexico	16.0	16.3	15.9	15.8	—
Paraguay	51.9	52.2	51.9	51.9	—
Peru	34.7	34.8	34.8	34.9	35.5
Poland	20.8	20.8	20.7	20.6	22.2
Russia	20.9	21.6	20.8	20.8	23.5
South Africa	8.5	8.6	8.5	8.7	8.5
Turkey	24.5	26.3	24.5	24.4	28.0
Uruguay	62.5	62.9	62.6	62.4	—

Notes: Column (1) repeats column (4) of Table 1. Columns (2) and (3) convert domestic-law stocks at the 31 December exchange rate and at the annual median of daily rates, instead of the annual average. Column (4) revalues foreign-law bonds at each year's exchange rate instead of keeping their dollar value at issue. Column (5) also removes holdings of public institutions. — : not available.

A.2 Duration differences

For bills and zero-coupon bonds, duration equals maturity. For fixed-coupon bonds we apply the formula of Section 2.3 to their coupons and principal, and for inflation-linked bonds we apply it in real terms, with the real coupon and the real yield. For floating-rate bonds, duration equals the time to the next coupon reset. The yield at issuance comes from official sources for domestic-law placements and from Dealogic for foreign-law placements. Duration for a floating-rate bond depends on how we treat its coupon resets, so maturity, which counts every bond over its full term, gives a measure free of that choice; Table 10 reports it, Table 11 reports the medians over a common window, and Table 12 reports the laws under which the foreign-law placements are written.

- **Maturity at issuance** (Table 10). Years from settlement to final payment, summarised as in Table 2.
- **Common window** (Table 11). Recomputes the medians with every country on 2006–2024.
- **Governing law** (Table 12). Splits foreign-law face by the law each placement is written under.

Table 10. Maturity at Issuance by Governing Law

	Domestic law				Foreign law				Gap	
	Median (1)	Mean (2)	Face (3)	GDP (4)	Median (5)	Mean (6)	Face (7)	GDP (8)	Median (9)	Face (10)
Brazil	4.59	7.29	4.45	4.36	10.40	16.41	16.24	16.46	5.81	11.79
Chile	9.84	13.81	10.67	10.81	11.42	16.48	16.83	16.60	1.58	6.16
Colombia	4.08	6.64	8.41	—	11.00	17.79	17.76	—	6.92	9.35
Hungary	3.85	4.94	3.08	2.91	8.50	9.39	10.82	10.64	4.65	7.74
Mexico	1.00	4.98	3.31	3.23	10.00	17.25	19.74	19.30	9.00	16.43
Paraguay	3.83	4.92	5.28	4.96	11.41	16.79	16.89	16.94	7.58	11.61
Peru	11.54	15.28	14.05	14.65	16.00	24.20	22.74	22.60	4.46	8.69
Poland	5.26	6.32	5.36	5.15	10.00	11.00	11.34	11.22	4.74	5.98
Russia	7.44	8.33	8.91	8.72	10.00	12.90	14.35	14.31	2.56	5.44
South Africa	9.62	11.34	6.72	6.84	12.00	16.91	17.18	16.90	2.38	10.46
Turkey	2.99	3.89	3.54	3.16	10.00	10.95	10.94	11.37	7.01	7.40
Uruguay	9.53	9.90	10.14	10.00	12.00	16.20	18.92	19.01	2.47	8.78

Notes: Years from settlement to final payment of each placement. Columns as in Table 2. — : not available.

Table 11. Maturity and Duration at Issuance: Common Window

	Maturity		Duration	
	Domestic (1)	Foreign (2)	Domestic (3)	Foreign (4)
Brazil	4.59	10.40	3.47	8.40
Chile	9.84	11.42	8.85	9.90
Colombia	4.08	11.00	3.75	8.71
Hungary	4.29	9.00	2.94	7.43
Mexico	1.00	10.00	0.48	8.87
Paraguay	3.83	11.41	3.46	8.74
Peru	11.54	16.00	8.64	11.77
Poland	5.32	10.00	2.60	8.15
Russia	7.57	10.00	5.62	8.00
South Africa	9.58	12.00	6.89	9.06
Turkey	4.25	10.00	1.64	7.68
Uruguay	9.53	12.00	8.19	9.45

Notes: Median maturity and median duration at issuance, in years, with every country on 2006–2024.

A.3 A simple model of the yield gap

- **Case 1. Payoffs in the investor's own currency.** The gap rises with the probability of default, and its slope reflects the difference in recoveries.

Table 12. Governing Law of Foreign-Law Placements

Country	Governing law (% of foreign-law face)
Brazil	NY 84, London 9, Tokyo 4, Frankfurt 2, Lux 2
Chile	NY 100
Colombia	NY 98, Tokyo 2
Hungary	London 72, NY 21, Tokyo 4, China 2, Frankfurt 1
Mexico	NY 90, Tokyo 8, Frankfurt 2, London 1
Paraguay	NY 100
Peru	NY 100
Poland	London 66, NY 27, Tokyo 5, Frankfurt 1, China 1
Russia	London 96, Frankfurt 4
South Africa	NY 90, London 5, Tokyo 3, Frankfurt 2
Turkey	NY 64, London 23, Frankfurt 7, Tokyo 6
Uruguay	NY 93, Tokyo 7

Notes: Share of foreign-law face issued under each law, from most to least. NY includes tranches that name New York beside the home law.

- **Case 2. Currency conversion with a forward.** The forward is written on the full face value and the bond pays only its recovery in default, so the bias depends on the probability of default. Includes the synthetic dollar bond, step by step.
- **Case 3. Both bonds in the same currency.** The estimate is a lower bound on the difference in recoveries, and the null hypothesis remains testable.
- **Multi-period extension.** The gap by maturity, and why time to maturity enters only as a control.

A.4 Depreciation at default

- δ from the pair of CDS contracts that settle in dollars and in local currency: same reference entity, same credit event, same auction, so δ is the ratio of the two spreads. Computed day by day, after dropping stale local-currency quotes.
- Why the gap carries $1/\delta$: the ordinary depreciation appears in both the price gap and the price level and cancels, over one period and over n periods.
- Which CDS to regress on: the gap multiplied by δ against the dollar CDS, or the raw gap against the local-currency CDS. Using the gap multiplied by δ against the local-currency CDS corrects twice.
- [Table: the slope of the gap multiplied by δ, for each country with a quoted local-

currency CDS.]

A.5 Instruments and curves

- Domestic-law curve sources by country.
- Brazil's day-count conventions.
- The Nelson–Siegel fit for Paraguay.
- Bond details, and the reasons each sample window starts and ends.

A.6 Stationarity, cointegration and inference

A.7 Robustness of the yield-gap estimates

- Net of Colombia's withholding tax.
- The liquidity control interacted with the CDS level.
- Alternative and common windows.
- Around rating actions.
- With the post-COVID data added back.
- Without Paraguay's most recent bond.

A.8 Binned scatters by country and bond-by-bond estimates

A.9 Placebo: Peru's Global Depositary Notes

- The instrument: a depositary bank's note over domestic-law Bonos Soberanos. New York law governs only the custody arrangement; the payment claim stays under Peruvian law.
- Market access: unrestricted for nonresidents, who hold 41 percent of the stock.
- The same construction and regression as Table 4, and the estimate.

A.10 The cost of issuing

Only five countries have bonds with a recorded fee under laws other than New York law, 50 bonds in all, and three quarters of them are yen bonds issued under Japanese law.

PROSPECTUS SUPPLEMENT
(TO PROSPECTUS DATED APRIL 15, 2021)

U.S. $2,000,000,000

Republic of Colombia

3.250% Global Bonds due 2032

The bonds will mature on April 22, 2032. The Republic of Colombia ("Colombia" or the "Republic") will pay interest on the bonds each April 22 and October 22, commencing on October 22, 2021. The bonds will be issued in denominations of U.S. $200,000 and integral multiples of U.S. $1,000 in excess thereof.

The bonds will be direct, general, unconditional, unsecured and unsubordinated external indebtedness of Colombia and will be backed by the full faith and credit of Colombia. The bonds will rank without any preference among themselves and equally with all other unsecured and unsubordinated external indebtedness of Colombia. It is understood that this provision shall not be construed so as to require Colombia to make payments under the bonds ratably with payments being made under any other external indebtedness.

Colombia may, at its option, redeem the bonds, in whole or in part, before maturity, on not less than 10 nor more than 60 days' notice on the terms described under "Description of the Bonds—Optional Redemption" in this prospectus supplement. The bonds will not be entitled to the benefit of any sinking fund.

The bonds will be issued under an indenture and constitute a separate series of debt securities under the indenture. The indenture contains provisions regarding future modifications to the terms of the bonds that differ from those applicable to Colombia's outstanding public external indebtedness issued prior to January 28, 2015. Under these provisions, which are described beginning on page 7 of the accompanying prospectus, Colombia may amend the payment provisions of any series of debt securities (including the bonds) and other reserve matters listed in the indenture with the consent of the holders of: (1) with respect to a single series of debt securities, more than 75% of the aggregate principal amount of the outstanding debt securities of such series; (2) with respect to two or more series of debt securities, if certain "uniformly applicable" requirements are met, more than 75% of the aggregate principal amount of the outstanding debt securities of all series affected by the proposed modification, taken in the aggregate; or (3) with respect to two or more series of debt securities, more than 66 2/3% of the aggregate principal amount of the outstanding bonds of all series affected by the proposed modification, taken in the aggregate, and more than 50% of the aggregate principal amount of the outstanding debt securities of each series affected by the proposed modification, taken individually.

Application will be made to list the bonds on the official list of the Luxembourg Stock Exchange and to trade them on the Euro MTF Market of the Luxembourg Stock Exchange.

See "Risk Factors" beginning on page S-7 to read about certain risks you should consider before investing in the bonds.

Section 309B(1)(c) of the Securities and Futures Act (Chapter 289 of Singapore) Notification

The bonds are prescribed capital markets products (as defined in the Securities and Futures (Capital Markets Products) Regulations 2018).

Neither the Securities and Exchange Commission, referred to as the SEC, nor any other regulatory body has approved or disapproved of these securities or determined if this prospectus supplement or the accompanying prospectus is truthful or complete. Any representation to the contrary is a criminal offense.

	Per bond	Total
Public offering price(1)	99.032%	U.S. $1,980,640,000
Underwriting discount	0.200%	U.S. $ 4,000,000
Proceeds, before expenses, to Colombia	98.832%	U.S. $1,976,640,000

(1) Purchasers will also be required to pay accrued interest, if any, from April 22, 2021, if settlement occurs after that date.

Source: Republic of Colombia, prospectus supplement (424B5) for the U.S.$2,000,000,000 3.250% Global Bonds due 2032, April 2021. SEC EDGAR accession 0001193125-21-124194, cover page. Highlight added.

Figure 5. The Cover of a Prospectus

Table 13. Issuer Expenses per Deal

Country	Prospectuses with an expense figure (1)	Expenses, median, US$ thousand (2)	Expenses, s.d., US$ thousand (3)	Share of the cost of a deal, % (4)
Brazil	29	185	121	5
Chile	30	150	106	20
Colombia	34	300	44	7
Hungary	7	175	71	5
Mexico	40	250	88	4
Peru	14	265	73	17
Poland	11	200	79	6
South Africa	14	305	187	12
Turkey	1	—	—	—
Uruguay	20	120	40	10

Notes: Expenses the issuer pays outside the underwriting fee (legal, listing, agents), as estimated in each SEC prospectus, New York-law bonds, 2006–2024. Where the stated figure includes the fee, the fee is subtracted. Column (4) is the median, deal by deal, of expenses over the fee plus expenses.

Table 14. Underwriting Fee on Bonds under Other Foreign Laws

	Bonds (1)	Median (2)	Mean (3)	Mean, face-weighted (4)
Hungary	6	30.0	33.3	22.0
Mexico	18	26.0	26.6	25.8
Poland	21	35.0	38.8	35.3
Turkey	1	13.0	13.0	13.0
Uruguay	4	8.0	8.0	8.0

Notes: As in Table 5, for bonds issued under Japanese, English and Swiss law, 2006–2024.

Table 15. Underwriting Fee by Governing Law and Market

	All countries			Same country: median		
Law and market	Bonds	Median	Mean	Mexico	Poland	Hungary
New York law	368	15.0	16.5	19.0	14.0	18.0
English law, euro	8	14.0	14.2	—	14.5	10.0
Japanese law, yen	37	29.0	29.2	26.0	35.0	30.0
Swiss franc	4	75.0	81.2	—	75.0	75.0
English law, hand-collected	37	14.0	13.5	—	14.0	12.5

Notes: Underwriting fee in basis points of face value, Dealogic, 2006–2024. The last three columns give the median fee of each country's bonds in each market. The last row uses English-law fees read from Luxembourg and London exchange filings.

Table 15 compares the fee by law and market. English-law bonds in euros pay about the same fee as New York-law bonds, 14 against 15 basis points, and in Poland 14.5 against 14.0. Yen bonds pay about twice as much, and Swiss-franc bonds more. Among these 50 bonds, a 30-year bond pays 5.5 basis points less than a 10-year bond of the same country and year, with a 95 percent interval from -8.8 to -2.1.

A.11 How often and how much governments issue

Governments borrow abroad less often than at home, and in larger operations. Table 16 reports, for each country, how often the government issues under each law and how much it raises each time, by quarter, over 2006–2024. The share of quarters with no foreign-law issuance ranges from 0.26 in Turkey to 0.83 in Paraguay. Under domestic law it is at most 0.25, and six countries issue domestic-law bonds every quarter. Relative to that quarter's GDP, the median foreign-law operation is larger than the median domestic-law one in ten of the twelve countries, with a ratio that ranges from 1.1 in Poland to 90 in Uruguay; Brazil and Turkey are the two exceptions.

We count operations, not bonds. An operation is one day on which bonds settle in a market, and we add up everything that settles that day. Re-openings of outstanding foreign-law bonds count as operations, because each is a new sale abroad with its own prospectus and fee. Both laws use the placements of the previous sections: Dealogic for foreign law and debt-office records for domestic law. We convert each operation to US dollars and divide it by that quarter's nominal GDP in US dollars.

A.12 Recovered debt in the numerical solution

We solve the model on a uniform grid for each bond. Domestic-law debt takes the values $b_D(i) = \bar{b}_D\, i/(N_D - 1)$ for $i \in \{0, 1, \ldots, N_D - 1\}$, and foreign-law debt is defined in the same way on N_F points. A government that regains access after a default carries $\nu_D b_D$ and $\nu_F b_F$. Because each grid is linear in its index and starts at zero, scaling the debt level by the recovery is the same as scaling the index: $\nu_D b_D(i) = b_D(\nu_D i)$. The recovered position $p_D = \nu_D i$ is generally not an integer, and the same holds for $p_F = \nu_F j$. Since recoveries are at most one, the recovered point never leaves the grid.

Each recovered position lies between two grid points, $i^- = \lfloor p_D \rfloor$ and $i^+ = \min(i^- + 1, N_D - 1)$, with weight $a = p_D - i^-$, and likewise j^-, j^+ and $c = p_F - j^-$ for foreign-law debt. We evaluate the new-issuance prices q_D^{ND} and q_F^{ND} and the continuation value at the

Table 16. Issuance by Governing Law, Quarterly

	Inaction rate		Operations per quarter		Spacing,	One operation, % of quarterly GDP		
	Foreign (1)	Domestic (2)	Foreign (3)	Domestic (4)	quarters (5)	Foreign (6)	Domestic (7)	Ratio (8)
Brazil	0.54	0.00	0.61	20.1	2.1	0.22	0.48	0.5
Chile	0.75	0.17	0.36	3.4	3.1	2.51	0.45	5.5
Colombia	0.55	0.09	0.50	8.4	2.3	1.95	0.28	7.0
Hungary	0.62	0.00	0.51	10.0	2.6	3.38	0.58	5.9
Mexico	0.34	0.00	0.84	15.1	1.5	0.61	0.25	2.4
Paraguay	0.83	0.25	0.17	1.7	3.7	5.11	0.15	34.6
Peru	0.80	0.05	0.22	6.7	4.4	2.54	0.09	26.8
Poland	0.41	0.00	0.99	5.4	1.7	0.92	0.81	1.1
Russia	0.80	0.01	0.20	8.6	3.1	0.61	0.09	7.0
South Africa	0.80	0.00	0.21	19.5	5.3	1.87	0.19	9.7
Turkey	0.26	0.00	1.12	5.6	1.4	0.73	0.76	1.0
Uruguay	0.64	0.19	0.39	4.0	2.9	7.95	0.09	90.0

Notes: 2006Q1–2024Q4. An operation is one settlement day in a market, with amounts summed. Foreign-law placements from Dealogic, counting re-openings as operations; domestic-law placements from debt-office records, bonds only. Exchange operations are excluded on both sides. Columns (1)–(2) give the share of quarters with no issuance, (3)–(4) the operations per quarter, (5) the mean number of quarters between foreign-law issuance quarters, (6)–(7) the median operation as a percentage of that quarter's GDP, and (8) the ratio of (6) to (7).

recovered point by bilinear interpolation of the four surrounding grid values,

$$\tilde{f}(p_D, p_F) = (1-a)(1-c)\, f[i^-, j^-] + a(1-c)\, f[i^+, j^-] + (1-a)c\, f[i^-, j^+] + ac\, f[i^+, j^+]. \tag{20}$$

The weights are nonnegative and sum to one, and when a recovered position falls on a grid point the formula uses that point alone. Each law's recovery therefore scales its own debt, and the recovered portfolio need not lie on the grid.

A.13 The slope we target

[Draft. The model has one currency, so its slope measures the recovery difference without depreciation at default. In the data, the same-currency gap carries the factor $1/\delta$ (Section 2.4, Appendix A.4), so the raw slope is a lower bound for the currency-free slope: 0.23 raw against 0.36 corrected for Colombia. Decide which slope the calibration targets. If the raw slope, state that the recovery gap and the welfare results of Section 5 are lower bounds.]

A.14 Robustness to the reentry probability

[Calibration under other values of θ, e.g. three and six years.]

A.15 The two experiments

Table 17. Same Durations, Different Recoveries: Foreign-Law Share (%)

	Duration of both bonds (years)				
ν_D / ν_F	1.00	2.00	3.75	5.50	8.71
0.200 / 0.400	100.0	100.0	100.0	100.0	100.0
0.400 / 0.420	100.0	100.0	100.0	100.0	100.0
0.400 / 0.471	100.0	100.0	100.0	100.0	100.0
0.400 / 0.550	100.0	100.0	100.0	100.0	100.0
0.400 / 0.700	100.0	100.0	100.0	100.0	100.0
0.471 / 0.400	0.0	0.0	0.0	0.0	0.0

Notes: Foreign-law share of debt when both bonds have the same duration. Other parameters at their calibrated values. The last row reverses which law has the higher recovery.

Table 18. Same Durations, Different Recoveries: Recovery Grid (%)

	ν_F				
ν_D	0.300	0.375	0.450	0.525	0.600
0.300	39.7	100.0	100.0	100.0	100.0
0.375	0.0	39.7	100.0	100.0	100.0
0.450	0.0	0.0	45.6	100.0	100.0
0.525	0.0	0.0	0.0	43.7	100.0
0.600	0.0	0.0	0.0	0.0	39.1

Notes: Foreign-law share of debt, both bonds with a duration of 3.75 years. On the diagonal the two bonds are identical and the split is not determined by the model.

A.16 The fee and the cost of borrowing

This appendix shows why a fee charged once per unit of face value issued weighs more on short bonds than on long ones.

Table 19. Same Recoveries, Different Durations: Foreign-Law Share (%)

	Duration, domestic / foreign law (years)		
$\nu_D = \nu_F$	1 / 8.71	2 / 8.71	3.75 / 8.71
0.00	47.8	2.0	0.3
0.40	58.8	2.0	0.3
0.4715	60.4	2.0	0.1
0.60	65.3	2.0	0.1
0.75	67.8	3.7	0.1

Notes: Foreign-law share of debt when both bonds have the same recovery. The last column uses the calibrated durations. Other parameters at their calibrated values.

What the fee is charged on. The fee f is charged on face value, the amount the government promises to repay, and it is paid once, when the bond is issued. This is how the prospectuses of Section 2.5 quote it: an underwriting discount of 0.200 percent on US$2 billion of face is US$4 million, whatever the price at which the bonds are sold.

The bond. A bond with decay λ and no coupon pays, per unit of face, λ next period, $\lambda(1-\lambda)$ the period after, and so on: each period a share λ of the face still outstanding matures. Its yield y is the constant rate that discounts these payments to the bond's price q:

$$q = \sum_{k=1}^{\infty} \frac{\lambda(1-\lambda)^{k-1}}{(1+y)^k} = \frac{\lambda}{1+y} \sum_{k=0}^{\infty} \left(\frac{1-\lambda}{1+y}\right)^k = \frac{\lambda}{1+y} \cdot \frac{1+y}{y+\lambda} = \frac{\lambda}{y+\lambda}. \tag{21}$$

Solving for the yield gives $y = \lambda/q - \lambda$. A one-period bond has $\lambda = 1$ and $y = 1/q - 1$. For a given yield, the price rises with λ: a bond that repays sooner is discounted less, so short bonds trade closer to par than long bonds.

The fee as a lower price. With the fee, the government sells one unit of face for q but keeps only $q - f$, while it owes the same payments. The yield it pays on the money it actually receives is therefore

$$\hat{y} = \frac{\lambda}{q-f} - \lambda. \tag{22}$$

Subtracting y and writing the difference over a common denominator,

$$\hat{y} - y = \lambda \left(\frac{1}{q-f} - \frac{1}{q}\right) = \frac{\lambda f}{q\,(q-f)} \;\approx\; \frac{\lambda f}{q^2}, \tag{23}$$

where the approximation replaces $q - f$ by q, since the fee is small relative to the price. The price enters twice. Dividing the fee by q turns a cost per unit of face into a cost per unit of money borrowed, and dividing by q again turns a change in the price into a change in the yield, since $\partial y / \partial q = -\lambda / q^2$.

Short against long bonds. We compare the wedge on two bonds of the same government that differ only in their decay rate and trade at the same yield. This is the relevant comparison in the model: the two foreign-law bonds share the same recovery, so they carry the same mean spread and the same mean yield. A shorter bond has a larger λ, which raises the wedge $\lambda f / q^2$, but also a higher price q, which lowers it. To see which force dominates, substitute the price from equation (21), $q = \lambda / (y + \lambda)$, into the wedge, so that it depends on λ and y alone:

$$\frac{\lambda f}{q^2} = \lambda f \cdot \frac{(y+\lambda)^2}{\lambda^2} = \frac{f\,(y+\lambda)^2}{\lambda}. \tag{24}$$

At a given yield, the wedge changes with the decay rate according to

$$\frac{\partial}{\partial \lambda}\left[\frac{f\,(y+\lambda)^2}{\lambda}\right] = f\,\frac{2(y+\lambda)\lambda - (y+\lambda)^2}{\lambda^2} = f\,\frac{(y+\lambda)(\lambda - y)}{\lambda^2}, \tag{25}$$

which is positive whenever $\lambda > y$, that is, whenever the bond repays a larger share of its face each period than its yield. Every bond in the paper satisfies this condition: the slowest decay rate we use is $\lambda_F = 0.099$, the foreign-law bond of Section 4.1 with a duration of 8.71 years, and yields are a few percent. The wedge is therefore larger on the shorter bond. Its price is higher, but the price falls with maturity much less than λ does, so even squared it offsets only part of the difference in λ.

The table evaluates these expressions at the fee of 25 basis points and a common yield of 4.2 percent, the risk-free rate plus Colombia's mean spread. The first two rows are the short and the long foreign-law bonds of Section 5.1; the third is the foreign-law bond of Section 4.1. To keep one unit of face outstanding, the government issues λ units each period and pays $f\lambda$ in fees; the last column is the wedge in the yield.

Bond	λ	Price q	$f\lambda$ (bp of face)	Wedge $\lambda f / q^2$ (bp a year)
One period	1.000	0.960	25.0	27.1
2.32 years	0.420	0.909	10.5	12.7
8.71 years	0.099	0.702	2.5	5.0

From one period to 8.71 years, λ falls by a factor of ten and the price by less than a third, so the wedge falls by a factor of more than five. A higher yield on the long bond would lower its price and raise its wedge, but by equation (24) the 8.71-year bond would need a yield above 23 percent, and the 2.32-year bond a yield above 25 percent, for its wedge to reach that of the one-period bond at 4.2 percent. [Add the mean prices of the two foreign-law bonds in the three-bond model and the implied wedges.]

A.17 The fee ladder

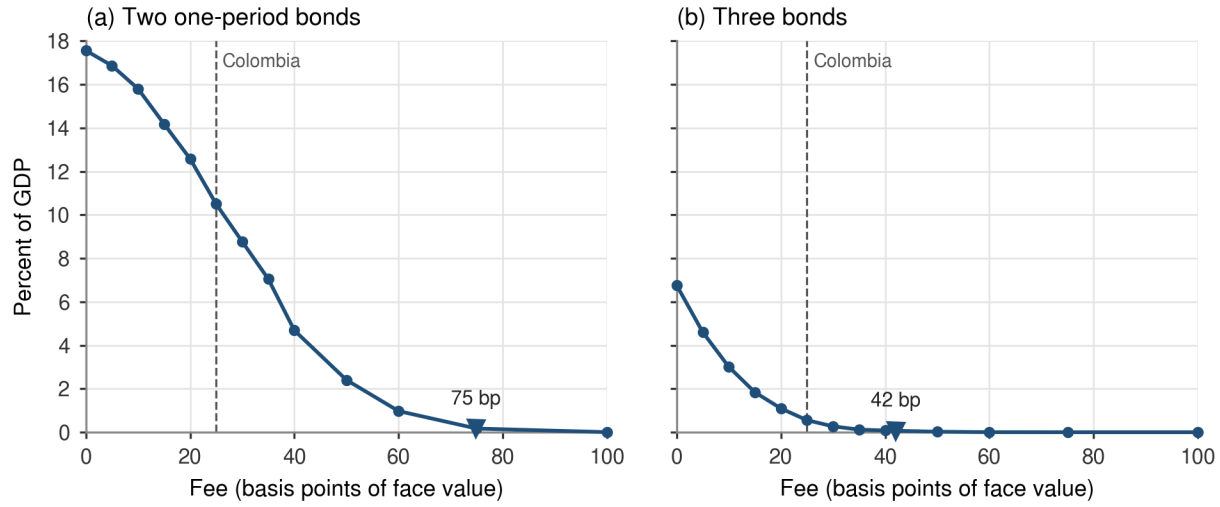


Figure 6. Foreign-Law Debt and the Fee

Notes: Mean debt in percent of GDP against the fee per unit of face value issued under foreign law. Panel (a): foreign-law debt when the government has a one-period domestic-law bond and a one-period foreign-law bond. Panel (b): short foreign-law debt when the government also has a foreign-law bond with a duration of 2.32 years. The dashed line marks Colombia's fee of 25 basis points. The triangle marks the fee at which the plotted debt falls below 1 percent of its value without a fee.

[Extend to the full range of fees, 0 to 300 basis points, with grid checks and the rows with the flat cost per issue; add the two-bond results at 25 and 35 basis points.]

A.18 Welfare: measurement and decomposition

Decomposition with policies fixed. The price of defaulted debt,

$$q_i^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\,\mathbb{E}_{y'|y}\left[q_i^D(y', b_D, b_F)\right] + \theta\,\nu_i\,q_i^{ND}(y, \nu_D b_D, \nu_F b_F)\,,$$

shows the two roles of the recovery: a higher ν_i raises the price investors pay, and it raises the debt the government carries at reentry. Holding the borrowing and default policies

Table 20. Shares of Debt and the Foreign-Law Fee

	Two bonds	Three bonds		
Fee (bp of face)	Foreign law	Short domestic	Short foreign	Long foreign
0	100.0	0.0	35.9	64.1
20	74.8	12.1	5.9	82.0
25 (Colombia)	[ ]	13.2	3.0	83.8
35	[ ]	18.5	0.7	80.7
Fee at which issuance stops (bp)				
Foreign law, two bonds	76			
Short foreign law, three bonds			35	

Notes: Shares of total face value, in percent, means over simulated periods. Two bonds: a one-period domestic-law bond and a one-period foreign-law bond. Three bonds: one-period domestic-law and foreign-law bonds and a foreign-law bond with a duration of 2.32 years. Issuance stops when the share falls below 1 percent. Other parameters as in Table 6. Appendix A.17 reports the full range of fees.

at their baseline values, we compute welfare in four economies: the baseline; new prices with the baseline debt at reentry; baseline prices with the new debt at reentry; and both changes together. The first comparison isolates the price effect and the second the reentry effect. The difference between the economy with $\nu_D = 0.47$, in which policies adjust, and the economy with both changes and fixed policies is the effect of the government's behaviour.

The value of market access. For an economy $\mathcal{E}$, let $\bar{V}^{\mathcal{E}}$ be the household's expected discounted utility along the equilibrium path, with consumption $y - \phi(y)$ while the government is excluded. Let $\bar{V}^{ND,\mathcal{E}}$ be the same value when exclusion carries no output loss, so that consumption equals y while excluded and the government still reenters owing $\nu_D b_D$ and $\nu_F b_F$. Let $\bar{V}(\bar{c}^{ND,\mathcal{E}}) = \sum_t \beta_H^t u(\bar{c}_t^{ND,\mathcal{E}})$ be the value of receiving for sure the mean path $\bar{c}_t^{ND,\mathcal{E}}$ of that consumption. Let $\bar{V}^A$ be the value of consuming output forever, and $\bar{V}^A(y^\infty)$ the value of consuming mean output $y^\infty = \sum_y \pi(y)\, y$ forever. Following Aguiar et al. (2020), the consumption-equivalent value of market access is

$$1+\chi^{\mathcal{E}} \equiv \left(\frac{\bar{V}^{\mathcal{E}}}{\bar{V}^{A}}\right)^{\frac{1}{1-\sigma}} = \underbrace{\left(\frac{\bar{V}^{\mathcal{E}}}{\bar{V}^{ND,\mathcal{E}}}\right)^{\frac{1}{1-\sigma}}}_{\text{default costs}} \underbrace{\left(\frac{\bar{V}^{ND,\mathcal{E}}}{\bar{V}(\bar{c}^{ND,\mathcal{E}})} \cdot \frac{\bar{V}^{A}(y^\infty)}{\bar{V}^{A}}\right)^{\frac{1}{1-\sigma}}}_{\text{variability}} \underbrace{\left(\frac{\bar{V}(\bar{c}^{ND,\mathcal{E}})}{\bar{V}^{A}(y^\infty)}\right)^{\frac{1}{1-\sigma}}}_{\text{front-loading}}. \quad (26)$$

The value of autarky is the same in the two economies, so the welfare gain of equation (19) is $1+\Delta = (1+\chi^{\mathcal{E}_1})/(1+\chi^{\mathcal{E}_0})$, and each term contributes the ratio of its own values across the two economies.

[Add the table with all four experiments ($\uparrow \nu_D, \downarrow \nu_F, \downarrow \nu_D, \uparrow \nu_F$) from the jamboree deck.]